

TDOT Slope Movement Monitoring System

East Tennessee Highway Corridor Risk Assessment

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Proof of Concept: I-40 and I-75 Mountain Corridors

Real Sentinel-1 InSAR, USGS Terrain, USGS Geology, and FHWA Traffic Data

GitHub: github.com/MominaLAli/tdot-slope-monitoring

Problem and Project Objective

TDOT problem

- Tennessee has many unstable slope sites across highway corridors.
- Annual manual inspection of every site is difficult at statewide scale.
- Terrain-only screening can miss flat areas that are actively moving.
- A decision-support tool is needed to combine movement, terrain, geology, and road exposure.

Proof-of-concept objective

- Detect real ground movement using Sentinel-1 InSAR.
- Extract terrain susceptibility features from USGS 3DEP DEM data.
- Integrate geology and traffic exposure into a risk score.
- Rank highway segments for inspection priority.
- Provide a dashboard that TDOT engineers can use without coding.

Key idea: use satellite displacement to identify moving slopes that terrain-only inspection may overlook.

Study Area and Data Sources

Corridors analyzed

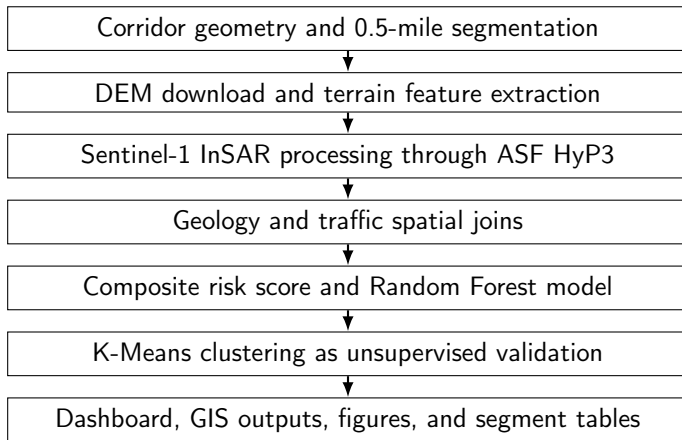
Road	Section	Length	Segments
I-40	Cookeville to North Carolina border	274 km	342
I-75	Jellico Mountain section, Campbell County	72 km	90
Total	East Tennessee	346 km	432

Input layers

Layer	Source and role
InSAR	Sentinel-1 via ASF HyP3; 35 pairs from Jan. 2022-Jan. 2024
Terrain	USGS 3DEP 30 m DEM; slope, roughness, curvature, relief
Geology	USGS Tennessee geology map; rock type per segment
Traffic	FHWA HPMS 2022; AADT and truck percentage

All layers are real and publicly available. The analysis uses 0.5-mile segment units and a 500 m buffer for spatial feature extraction.

Processing Pipeline



The pipeline converts raw geospatial and satellite products into ranked segment-level inspection priorities.

Risk Score Formulation

Composite risk score

$$\text{Risk} = 0.35P_{ML} + 0.25S_{\text{terrain}} + 0.20S_{InSAR} \\ + 0.12S_{\text{road}} + 0.08S_{\text{geo}}$$

Component	Meaning
P_{ML}	Random Forest probability of slope-risk class
S_{terrain}	Mean slope, elevation range, and terrain roughness
S_{InSAR}	Mean displacement, displacement trend, and displacement variability
S_{road}	AADT and truck volume exposure
S_{geo}	Rock-type susceptibility score

All continuous variables are min-max normalized before the weighted score is calculated.

Risk Category Results

Risk Category	Score Range	I-40	I-75	Total
Low	0.00-0.25	242	53	295
Moderate	0.25-0.50	34	10	44
High	0.50-0.75	66	25	91
Critical	0.75-1.00	0	2	2
Total		342	90	432

- Most segments are Low risk, but 93 segments are High or Critical.
- The two Critical segments are both located on the I-75 Jellico section.
- The category thresholds provide an inspection-priority ranking rather than a final engineering diagnosis.

Machine Learning Model Performance

Model comparison

Model	5-fold CV ROC-AUC
Random Forest, I-40 only	0.952 ± 0.055
Random Forest, combined corridors	0.945 ± 0.050
Gradient Boosting	0.986 ± 0.006
Logistic Regression	0.997 ± 0.002

Interpretation note High AUC values reflect proxy labels derived from the same feature space. These results are useful for proof-of-concept testing, but they should be recalibrated after TDOT USMP inspection points are available.

Primary model Random Forest was selected as the main model because it handles mixed data types, nonlinear relationships, and feature-importance estimation.

Unsupervised Validation

K-Means cluster profiles

Cluster	Risk	N	Mean slope
0	Moderate	149	5.2°
1	Low	128	4.1°
2	High	65	13.7°

- Cluster 0: displacement-driven moderate risk
- Cluster 1: flat-terrain low risk
- Cluster 2: steep-terrain high risk

Why this is important K-Means was applied directly to real feature measurements with no proxy labels. The clustering results are consistent with the supervised model, suggesting that the terrain and InSAR signals are meaningful before USMP ground-truth validation.

- Optimal K: 3
- Silhouette score: 0.376
- PCA variance explained: 67.2% in first two components

Key Finding: Segment 149 on I-40

Segment profile

Property	Value
Location	I-40, east of Knoxville
Mean slope	5.2°
Rock class	Carbonate limestone
Valid InSAR pairs	32 of 35
Mean displacement	-3.318 mm
Trend	-1.785 mm/month
Cumulative movement	Approx. -43 mm over two years
Mean coherence	0.420

Why this matters

- The segment is relatively flat, so terrain-only screening classifies it as low risk.
- Real Sentinel-1 InSAR indicates sustained subsidence over two years.
- The integrated model elevates the segment to high risk.

Segment 149 demonstrates the central value of InSAR: it can identify movement that terrain-only inspection would likely miss.

Additional Findings

I-75 Jellico critical segments

- Segments 1023 and 1024 in Campbell County are classified as Critical.
- Risk scores are 0.797 and 0.800.
- Both segments have clastic sedimentary rock and slopes near 15.5°.
- Findings are consistent with known Jellico-area slope concerns.

Geology and risk

Rock class	N	High/Critical
Metamorphic/crystalline	1	1
Clastic sedimentary	186	91
Other	70	2
Carbonate	175	0

Clastic sedimentary units account for 91 of 93

High/Critical segments, making geology a major risk discriminator in the current study area.

Dashboard

Six dashboard tabs

- **Risk Map:** interactive segment risk map
- **Analysis:** risk profiles and feature importance
- **Geology and InSAR:** rock type, coherence, displacement
- **Corridor Compare:** I-40 versus I-75 metrics
- **Segment Table:** filterable table and CSV export
- **Risk Clustering:** K-Means PCA and Segment 149 analysis

Run command: `streamlit run streamlit_app.py`

Deliverables

Main project outputs

- GIS-ready risk segment GeoJSON
- Standalone interactive HTML map
- Top-risk segment table for inspection priority
- Feature-importance table for model interpretation
- Unsupervised clustering figure
- Segment 149 deep-dive figure

These outputs support engineering review and inspection prioritization. They do not replace field inspection or geotechnical judgment.

Limitations and Next Steps

Current limitations

- Proxy labels are not a substitute for TDOT USMP ground truth.
- I-75 InSAR processing is pending.
- Standard k-fold cross-validation may overestimate performance for spatial data.
- 30 m DEM data may miss fine-scale slope features.

Recommended next steps

1. Obtain TDOT USMP inspection points for validation.
2. Submit I-75 ASF HyP3 InSAR jobs.
3. Implement spatial block cross-validation.
4. Add higher-resolution Tennessee LiDAR data.
5. Extend to I-81, US-441, and US-64.

Conclusion

- The project demonstrates an end-to-end slope monitoring workflow for TDOT using real public datasets.
- The system covers 432 highway segments across 346 km of East Tennessee corridors.
- Segment 149 shows that InSAR can detect sustained movement on terrain that would otherwise appear low risk.
- I-75 Jellico results show that the method also identifies steep, geologically susceptible critical segments.
- The most important next step is ground-truth validation with TDOT USMP data.

InSAR-integrated monitoring can help prioritize inspection by detecting both visible terrain risk and hidden ground movement.